

Measurement choices in longitudinal citation language comparisons around replication projects

Dongdong Guo¹ and Jiaxuan Li^{1*}

¹Shanxi College of Applied Science and Technology, Taiyuan, China.

*Corresponding author(s). E-mail(s): jiaxuanli0817@gmail.com;
Contributing authors: 835741464@qq.com;

Abstract

Longitudinal citation-language studies depend on both the indicators measured and the texts selected for analysis. We examine these choices using 105 source papers from four replication projects in psychology and experimental economics. The archive contains 15,280 text units, deduplicated within each source; an empirical-restatement filter retains 8,427. We compare three fixed literal-word families and retain an archived model-based wording score as a secondary measure. Papers are grouped by selected replication criteria, and changes are aligned to project publication years. On the same 72 eligible papers, filtering changes the between-group contrast in pre-post replication-word incidence from -0.22 to -1.77 percentage points. The paired shift is -1.55 points (conditional 95% source-bootstrap interval -3.03 to -0.24). Artificial retention matched on archived year reproduces -0.50 points; matching year and length reproduces -0.65 points. The observed shift lies near the simulation-envelope boundary, leaving a descriptive -0.90 -point difference from the reference expectation. Adverb and modal results lie inside every reference envelope. Omitting the largest project or narrowing the time window yields filter-shift intervals that include zero. The model-score contrast is also imprecise, with a mixed rubric and reused development data limiting interpretation. This exploratory case shows how selection and text composition alter longitudinal descriptions in a psychology-dominated archive. Reporting retained and excluded populations together makes those dependencies visible; semantic acknowledgment, belief updating and causal responses to replication require separate evidence.

Keywords: Citation context, measurement, replication, text selection, scientific communication, bibliometrics

047 Introduction

048
049 Citation-language studies ask what citations communicate beyond the attention
050 recorded by citation counts. Content-based citation analysis examines the surrounding
051 language and relationships between citing and cited work (Ding et al., 2014). Citing
052 texts serve different functions, so their wording, location and relation to the cited work
053 matter for interpretation (Tahamtan and Bornmann, 2019; Budi and Yaniasih, 2023).
054 Studies of hedging and citation concepts have used those features to examine how
055 findings and ideas circulate (Small et al., 2019; Bornmann et al., 2020). Longitudinal
056 comparisons add a measurement problem: an apparent change can depend on which
057 texts enter the comparison as well as on the indicator measured in them.

058 Citation function can be annotated as the reason for citing a work and predicted
059 using linguistic features (Teufel et al., 2006). Later resources and models include SciCite,
060 which couples intent labels with structural information (Cohan et al., 2019); Kunnath
061 et al. (2021) review the classification schemes, contextual features and evaluation
062 choices across this literature. Recent approaches make the importance of representation
063 explicit: Ghosal et al. (2024) incorporate information about the cited paper, while
064 Jiang (2025) compare representations of in-text citations, sentences and surrounding
065 contexts. These approaches improve the information available for classification; they do
066 not remove the need to define the population over which a classified label is aggregated.

067 The semantic unit is also consequential. At the annotation stage, Athar and Teufel
068 (2012) show that sentence-only contexts can miss negative sentiment, and Taşkın
069 (2026) report that disagreement about citation-text boundaries reduces classification
070 reliability. MultiCite represents contexts that can span several sentences and express
071 several intents (Lauscher et al., 2022). The scite system distinguishes supporting,
072 contrasting and mentioning statements (Nicholson et al., 2021). These labels address
073 a semantic relationship, whereas an empirical-restatement filter decides which texts
074 enter an analysis. The question here concerns that later aggregation step: once a fixed
075 filter retains some texts and excludes others, how does the choice alter a longitudinal
076 contrast calculated on the same source papers?

077 Replication-linked papers provide a concrete case. New evidence could affect the
078 attention an earlier paper receives or the way its findings are described. Citation-
079 count studies examine the former (Schafmeister, 2021; Serra-Garcia and Gneezy, 2021;
080 Clark et al., 2023); Hardwicke et al. (2021) examine citation content after strongly
081 contradictory replications. Restricting a corpus to apparent empirical restatements
082 may exclude contextual or critical discussion. If retention also differs across years or
083 text lengths, a filtered comparison can reflect that changing composition. The source
084 paper itself is an imperfect semantic unit: later authors may cite different findings from
085 the same study (Leng, 2021). A replication criterion attached to one result therefore
086 cannot determine what every citation to that paper endorses.

087 Indicator definition creates a second distinction. Specified word forms, rhetori-
088 cal commitment and acknowledgment of contrary evidence are different observables.
089 The word replication does not by itself establish acknowledgment of an unsuccessful
090 replication; possibly may occur in quotation, negation or an unrelated proposition.
091 Commitment and certainty require explicit definitions (Hyland, 1998; Pei and Jurgens,
092 2021). More generally, transparent operational definitions are needed to assess whether

a numerical measure supports its intended interpretation (Flake and Fried, 2020);
 repeatable model outputs need not be accurate semantic classifications (Nishikawa
 and Koshiba, 2024). We consequently keep literal incidence separate from an inherited
 model-based wording score and from claims about author belief.

We examine these choices in a fixed archive from four replication projects, mainly
 in psychology and also in experimental economics. The analysis asks three linked
 questions: how retained and excluded texts differ in literal-word incidence; how filtering
 changes a pre–post criterion-group contrast when the source papers are held fixed;
 and how much of that shift is reproduced by artificial retention matched on archived
 year and text length. An exact decomposition describes the selection gap, and frozen
 composition benchmarks provide a reference for the observed shift. The contribution
 is an inspectable account of selection sensitivity in this archive, with all word families
 and adverse specifications retained, rather than a new citation classifier or a test of
 scientific self-correction.

Methods

Study sequence and scope

This exploratory reanalysis uses an inherited archive from four purposively selected
 replication projects. The original model-score results preceded the fixed literal rules;
 matched lexical findings preceded the decomposition, influence analyses and frozen
 retention benchmark. We retain all specified word families and unfavorable sensitivities.
 Analysis freezes document this sequence but do not constitute preregistration. We
 collected no new human ratings or model-generated substitutes for them.

Source papers and archived texts

The archive contains 135 distinct-title records derived from 141 selected targets in
 public follow-up data (Altmejd et al., 2019). Two projects are the Reproducibility
 Project: Psychology (RPP; Open Science Collaboration, 2015) and the Experimental
 Economics Replication Project (EERP; Camerer et al., 2016). The others are Many
 Labs 1 (Klein et al., 2014) and Many Labs 3 (Ebersole et al., 2016). Title-level reduction
 merged or omitted additional targets within papers, so the archive is not a project
 census. The historical resolver queried Semantic Scholar by title, retained candidates
 within two publication years of the source record, and selected the most-cited candidate.
 A legacy mismatch screen left candidates with fewer than 20 citations unresolved.
 This heuristic can exclude genuine low-cited papers and does not establish identity. Of
 135 title records, 25 remained unresolved; auditing the 110 returned candidates and
 excluding two wrong-paper matches left 108 source papers. Similar resolution rates
 across groups do not establish unbiased coverage. Online Resource 1 gives the full
 selection flow and source adjudications.

Criterion-positive denotes a reported or reproducibly reconstructed selected replica-
 tion test with nominal $p < .05$ and an effect in the original direction or pattern, using
 the project aggregate for multi-laboratory studies and all components of an explicit
 joint rule. Other resolved targets are criterion-negative. The retained conservative

baseline has 105 papers: 48 positive and 57 negative (Table 1). Three interpretation-dependent cases remain separate. A subsequent check resolved the archived rpp.73 Staunton statistic as positive without uniquely determining cross-site aggregation; we report a named 106-source sensitivity. For rpp.154 the mean is recoverable but a primary nominal p-value is inapplicable and historical interval rules conflict. Rpp.19 has target/archive inconsistencies. These source-associated criteria are not labels for every result or citation of a paper.

We used the Semantic Scholar Academic Graph citation endpoint to link each original paper to citing papers and retrieve stored citation contexts and publication years (Kinney et al., 2023). Retrieval took place in July 2026. Requests contained 100 citation records and targeted approximately 300 dated contexts per source paper. For papers with no more than 1,500 citations, offsets were 0, 100 and 200. For more highly cited papers, three offsets were placed deterministically and evenly from zero to the smaller of citation count minus 100 and 9,000. This procedure spread coverage across the accessible history but was not a uniform random sample of citations. The 9,000 starting-offset cap was the archived implementation’s conservative bound within a 10,000-record window, not a newly verified current API guarantee. The retrieval code retained contexts whose original length exceeded 40 characters, then stripped whitespace and deduplicated by citing year and the first 80 stripped characters. This initial prefix rule may merge distinct continuations.

After the two source-paper exclusions, the archive contains 15,544 retrieved records. Nine lack matching filter outputs and remain unclassified. Exact-text deduplication within each source retains the earliest archived year, removing 144 of the 15,535 scored records. The resulting 15,391 text units include 8,480 filter-positive units. Restriction to the 105 baseline papers leaves 15,280 units and 8,427 positives. The cache preserves text and year but lacks complete citing-paper and version identifiers. We therefore count text units deduplicated within source, rather than independently verified citing articles. Conflicting dates remain documented, and this later deduplication cannot recover contexts removed by the initial prefix rule.

Figure 1 connects source-paper reconstruction, text accounting and frozen selection to the matched comparison. The analysis fixes both source support and the all-text denominator before comparing retained and excluded texts.

Frozen text selection and literal observables

The archived binary filter used Qwen2.5-72B-Instruct at temperature zero. Its instruction selected empirical restatements and excluded method, task, scale, dataset, theoretical-description and generic-background uses. We call retained and excluded texts Qwen-positive and Qwen-negative, respectively. These fixed model decisions are distinct from the criterion-positive and criterion-negative source groups and have not been independently verified as assertion labels. Both text strata enter the lexical analyses; excluded texts receive no wording score, rather than a score of zero.

The first literal family comprises possibly, perhaps and probably, selected as published cue examples rather than a complete hedge lexicon (Kilicoglu and Bergler, 2008). The second comprises replicate, replicates, replicated, replicating, replication and replications. Originally lowercase may, might and could form a polysemy-sensitive

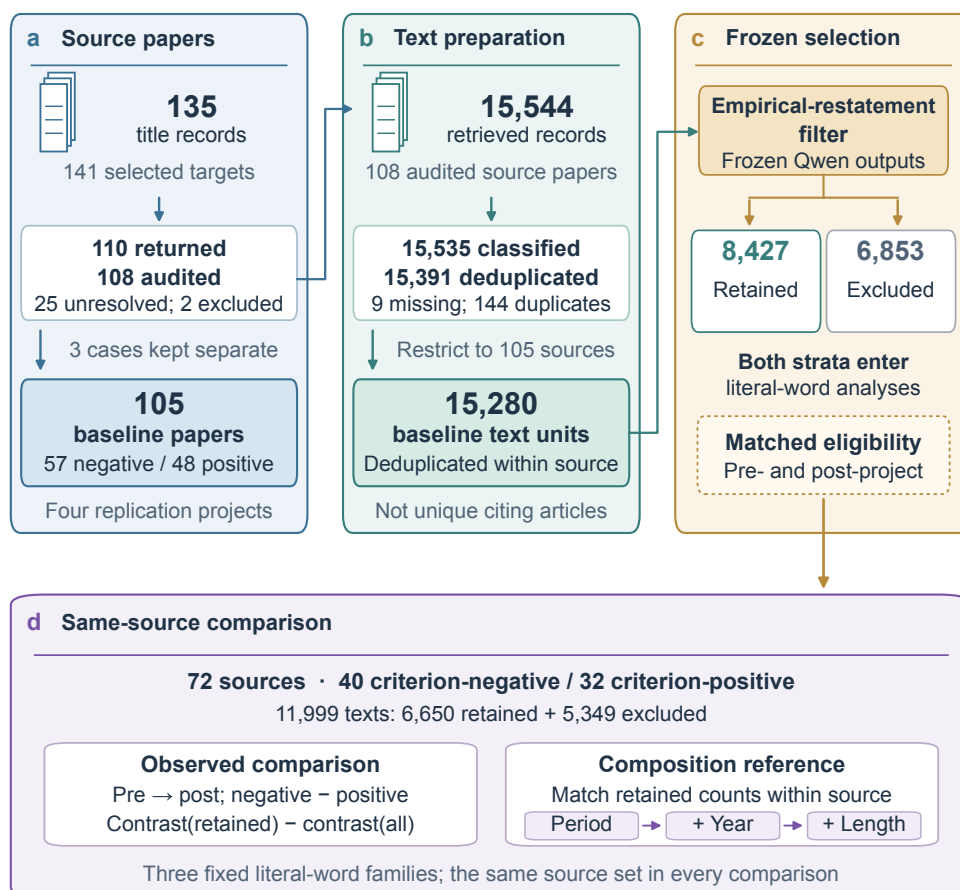


Fig. 1 Population reconstruction and matched comparison design. **a** Source reconstruction excludes two wrong-paper matches; three interpretation-dependent cases remain separately documented. **b** Accounting for archived texts removes nine records without filter outputs and 144 within-source duplicates before restriction to the baseline. **c** Frozen Qwen outputs define retained and excluded strata, distinct from source replication criteria. Matched eligibility requires at least ten all-text and ten retained units in each period; 33 baseline sources do not qualify. **d** Observed comparisons and nested composition references use the same 72 sources. Reference subsets preserve retained counts within source and period, then within source and archived year, then also length band. Boxes are schematic; text units are not independently verified citing articles

check. The rules fix Unicode token boundaries, case handling, an ASCII numeric-adjacency guard for modals and complete-token prefix checks. No stemming, scope analysis or semantic disambiguation is applied. Each text contributes a binary any-hit indicator per family. Every match is retained, including other senses, quotation and unrelated targets. Full rules and all twelve wordwise counts are in Online Resource 1.

231 Project-aligned comparisons and exact selection decomposition

232 Project journal-publication years provide transparent alignment conventions: RPP
233 2015, EERP 2016, Many Labs 1 2014 and Many Labs 3 2016. They do not measure
234 citing authors' awareness. All-text lexical comparisons require at least ten texts before
235 and ten at or after the project year, retaining sources with no word hits. Matched
236 comparisons additionally require at least ten Qwen-positive texts in each period and
237 use the identical source set for all-text and selected-text rates. Each source is weighted
238 equally. For each rate, we calculate post-minus-pre change, then the criterion-negative
239 mean minus the criterion-positive mean. Project results concern both groups; neither
240 is an untreated control.

241 For a source and period, let p^+ and p^- be literal-family incidence in Qwen-positive
242 and Qwen-negative texts, q the excluded fraction, and $d = p^+ - p^-$. The selected-minus-
243 all incidence gap is exactly $G = qd$. Its pre-post change has the symmetric identity
244 $\Delta G = \Delta q (d_{\text{post}} + d_{\text{pre}})/2 + \Delta d (q_{\text{post}} + q_{\text{pre}})/2$. We label the terms the retention-share
245 and stratum-distribution components, respectively. Here q is the excluded share, so
246 the first term reflects changing retention through its complement. Components are
247 computed within each source before criterion-group averaging. They are algebraic
248 allocations rather than causal pathways or percentages explained. Undefined excluded-
249 stratum incidence would remain missing; every source in the main matched set has
250 excluded texts in both periods.

251 We retain every single-source and leave-project-out result, within-project estimates,
252 equal-project and observed-project-share standardizations, and a sensitivity requiring
253 ten excluded texts per period. Small project cells do not support broad inference;
254 intervals are suppressed where resampling would keep a singleton criterion cell fixed.
255 Timing alternatives use the recorded manuscript-upload convention, exclusion of the
256 event year, symmetric five-year windows without that year, and the latest recorded
257 text year. Their eligible populations can differ.

259 Matched artificial-retention benchmarks

260 On the fixed 72-source matched set, we compare actual retention with uniform artificial
261 subsets preserving the observed retained count within progressively finer strata: source
262 $\times$ period, source $\times$ archived year, and source $\times$ archived year $\times$ pre-existing character-
263 length bands (0–69, 70–350, 351–700, 701–1,200 and at least 1,201). The schemes
264 retain all three literal families, 11,999 texts and 6,650 selected positions. We calculate
265 exact marginal expectations and simulate 19,999 joint three-family selections using
266 PCG64 seed 2026100602. Central 95% simulation envelopes describe these stipulated
267 finite-corpus schemes; they are not source-bootstrap confidence intervals or causal null
268 distributions. We report all nine family-by-scheme results and numerical precision of
269 the envelope endpoints in Online Resource 1.

272 Archived model scores and measurement agreement

273 GPT-5.6 Luna and the SiliconFlow deepseek-ai/DeepSeek-V4-Flash endpoint rated
274 Qwen-positive texts using five verbal anchors: 0, fully tentative or strongly conditional;
275 0.25, cautious suggestion; 0.5, neutral or distanced report; 0.75, strong evidential
276

statement; and 1, categorical, causal, axiomatic or unqualified general-present assertion. Explicit failed-replication or conflicting-evidence discussion was normally capped at 0.25, with categorical rejection allowed a score of 1. The legacy human codebook used the same exception. The exact unrounded mean of both scores is therefore a policy-defined mixture of rhetorical commitment and evidential qualification, rather than a monotonic measure of endorsement of one proposition. Inputs omitted source criterion, project, year, human rating and a validated target proposition.

On 248 ordinary paired-positive development items, the revised equal-weight score correlates with human means at Spearman $\rho = 0.695$ and Pearson $r = 0.707$. It meets the archived model-human rank-agreement criterion of 0.60 and improves on the former scorer (Online Resource 1). These items informed both development and evaluation, so the result is not independent validation. Human agreement is assertion $\kappa = 0.653$ across 330 ordinary items and ordinal certainty $\alpha = 0.562$ across the 248 paired-positive items. Certainty agreement remains below its separate internal criterion. Online Resource 1 reports score compression, error-by-period diagnostics and production repeatability, alongside the historical exploratory 150-item acknowledgment task. These checks describe the scoring procedure without validating target-specific updating.

The level endpoint requires at least 15 Qwen-positive texts per source. Lifetime drift uses at least three adaptive bins of at least eight positive texts each, at least three distinct rounded bin-mean years and a span of four years. A text-count-weighted regression of $\log(1 - \text{bin-mean score})$ on year has slope γ ; transformed drift is $1 - \exp(\gamma)$. The accompanying raw-score slope is distinct. Project-aligned score change uses the same ten-positive-texts-per-period rule as the matched lexical comparison. These endpoints have different support and are descriptive rather than causal.

Uncertainty and computational verification

Intervals are conditional percentile summaries from 19,999 source-paper bootstrap draws within criterion groups (Efron, 2003), with paired quantities resampled jointly. They hold retrieval, text, source criteria, dates, rules and model outputs fixed. Original lexical and model-score intervals retain seed 20261005; the later decomposition uses its frozen seed 20261006. Small interval-endpoint differences between these streams are documented, not substituted silently. Pointwise intervals do not provide a multiplicity-adjusted confirmatory family. Resampling known exact-text-sharing components checks only the dependence visible in identical strings; citing-article dependence is not fully observable. Four purposive projects do not support population-level inference over scientific fields.

Separate implementations check source-level identities, component sums, selections and uncertainty calculations; this verification concerns numerical agreement. Generative models assisted implementation, drafting and critique, and the human authors remain responsible for the manuscript and declarations. Versioned supplementary files contain the inputs and code for these checks. A separate pilot links existing filter outputs to independently produced article-level valence judgments. Its recovery rules, denominators and descriptive results are in Online Resource 1, Section S12, and Online Resource 4.

Results

Corpus support and observable wording

The baseline contained 15,280 text units from 105 source papers, comprising 8,427 Qwen-positive and 6,853 Qwen-negative units. Source support was uneven (Table 1). Equal-paper retention fractions were 0.5960 in criterion-negative papers and 0.5168 in criterion-positive papers. Their difference was 0.0792 (conditional 95% interval 0.0321–0.1256), describing filter decisions rather than independently measured assertion prevalence.

Table 1 Source-associated criteria and retained baseline. These are source papers, not independent citing articles.

Project	Retained papers	Positive	Negative	Separate	Baseline
RPP	74	27	44	3	71
EERP	14	9	5	0	14
Many Labs 1	11	9	2	0	11
Many Labs 3	9	3	6	0	9
Total	108	48	57	3	105

The three adverbs occurred in 87 texts (0.57%), replication-word forms in 304 (1.99%), and lowercase modals in 1,557 (10.19%). The corresponding Qwen-positive counts were 62, 133 and 1,124 (Fig. 2). The filter therefore excluded 171 texts containing replication-word forms from the scored subset. Those exclusions cannot be counted as errors or missed acknowledgments without semantic annotation. The three-adverb family was sparse and covers only a small part of uncertainty vocabulary.

All-text project-aligned comparisons retain 85 sources, 44 criterion-negative and 41 positive. Equal-source post-minus-pre incidence contrasts are -0.23 percentage points (95% interval -1.13 to $+0.67$) for the three adverbs, -0.68 (-2.76 to $+1.21$) for replication forms and $+2.57$ (-1.40 to $+6.86$) for lowercase modals. All include zero. This all-text population differs from the matched 72-source population and from the 85-source lifetime-score endpoint.

The same-source comparison changes under filtering

The matched set contained 72 papers, 40 criterion-negative and 32 positive, with 4,812 pre-period and 7,187 post-period text units. The replication-word change contrast was -0.22 percentage points using all texts and -1.77 using Qwen-positive texts. The paired filtered-minus-all shift was -1.55 points (original pointwise interval -3.03 to -0.24), although both individual contrast intervals included zero. Shifts were -0.08

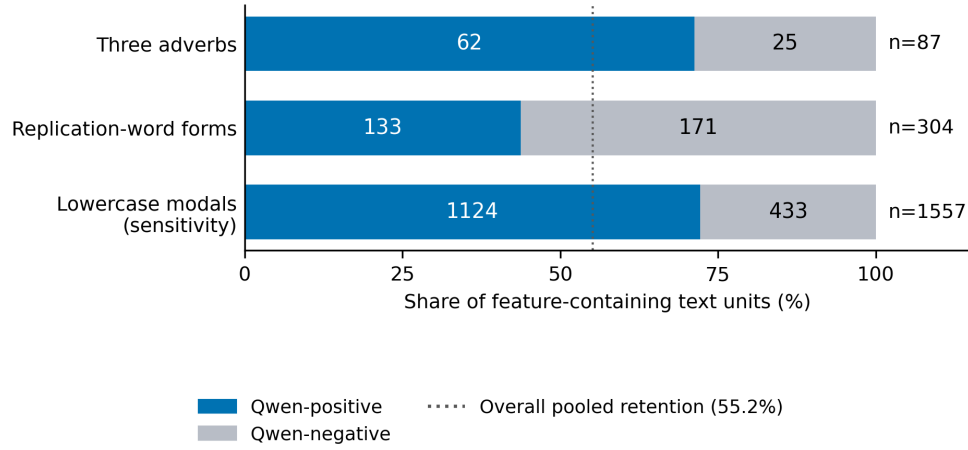


Fig. 2 Filter decisions among texts containing each literal family in the 105-source baseline. Bar widths show within-family percentages; labels give pooled text counts. The dotted reference is the overall pooled retention fraction. Families may overlap. These counts describe string presence and model outputs, without semantic classification

points (-0.72 to $+0.52$) for the three adverbs and -1.09 (-3.33 to $+1.31$) for modal forms (Fig. 3).

For replication forms, the exact decomposition allocated -0.23 points to the retention-share term and -1.32 points to the stratum-distribution term (Table 2). Within criterion-negative sources, equal-paper incidence rose from 2.09% to 2.45% in all texts but fell from 2.69% to 1.52% in retained texts. Incidence in excluded texts rose from 1.37% to 3.21%. These source-weighted descriptions account for the observed difference without identifying causal mechanisms or shares of a causal effect.

Table 2 Symmetric algebraic allocation of the matched filter shift. Entries are percentage points with conditional 95% source-bootstrap intervals from the extension stream, seed 20261006. Total point estimates reproduce Fig. 3; small interval-endpoint differences reflect the distinct fixed random stream.

Observable	Total shift	Retention-share term	Stratum-distribution term
Three adverbs	-0.076 [-0.717, +0.532]	+0.074 [+0.003, +0.154]	-0.150 [-0.802, +0.455]
Replication-word forms	-1.553 [-3.044, -0.208]	-0.233 [-0.410, -0.077]	-1.320 [-2.687, -0.064]
Lowercase modals (sensitivity)	-1.093 [-3.362, +1.344]	+0.168 [-0.221, +0.581]	-1.261 [-3.513, +1.182]

No single-source deletion reversed the replication-word shift; estimates ranged from -1.71 to -1.19 points. RPP nevertheless supplied 51 of the 72 sources. Omitting it

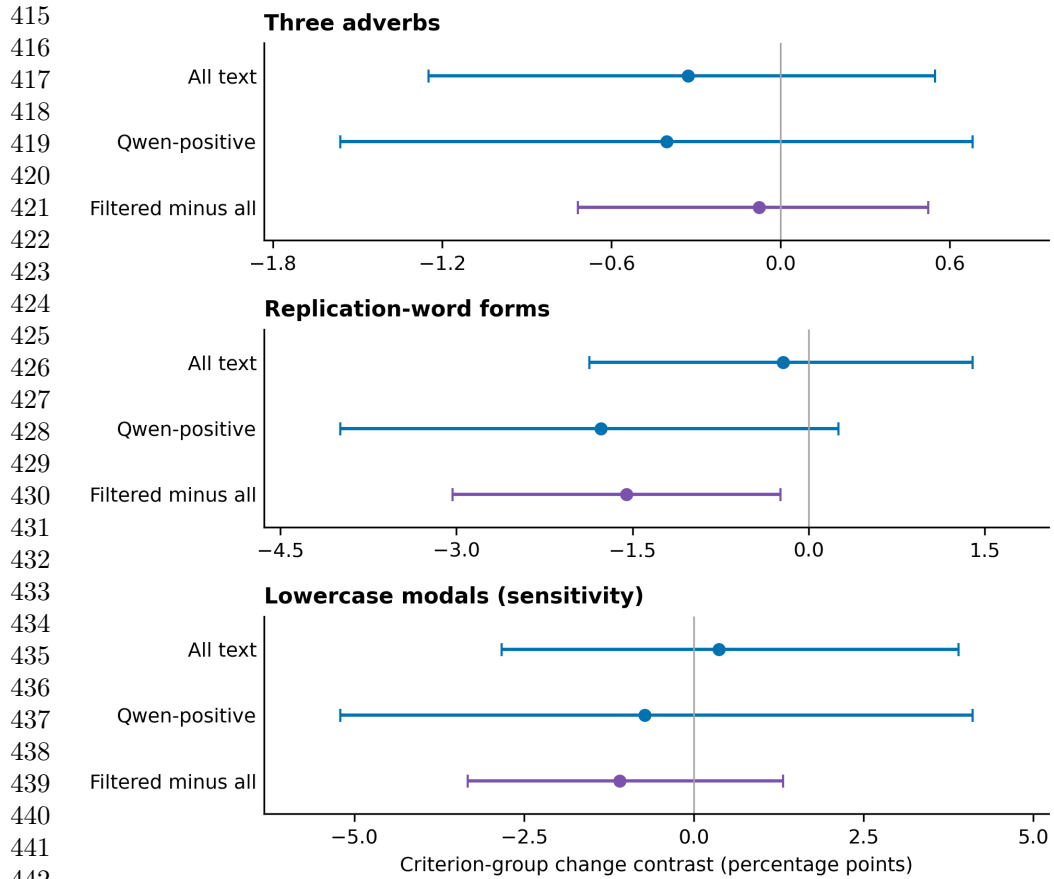


Fig. 3 All predefined families on the identical 72 sources. The first two rows are criterion-negative-minus-positive differences in post-minus-pre incidence; the last row is their paired filtered-minus-all difference. All values are percentage points; horizontal scales differ. Bars retain the original 19,999-draw pointwise source-bootstrap intervals, seed 20261005. A filter shift is not a semantic-error rate or a causal replication effect

reduced the shift to -0.70 points (extension interval -2.39 to $+0.90$). Requiring ten excluded texts per period retained 59 sources and yielded -1.44 points (-3.13 to $+0.04$). A symmetric five-year window excluding the project year retained 61 sources and yielded -1.52 points (-3.46 to $+0.13$). All three intervals included zero. Equal-project and observed-project-share standardizations yielded -1.02 and -1.72 points, respectively. We do not summarize their small-cell uncertainty with a headline interval. Online Resource 1 retains the full project, timing and influence results.

Resampling components linked by identical text preserved the replication-form interval below zero, while leaving dependence through unidentified citing articles unresolved. Under this resampling, the small positive adverb retention-share term no longer had an interval excluding zero. Both modal component intervals included zero, limiting support for a pattern shared across all three families.

Year and length composition reproduce part of the filtering contrast

For replication-word forms, expected artificial-retention shifts were 0.00 percentage points after matching source-period counts, -0.50 after matching archived year, and -0.65 after also matching length. The observed -1.55 -point shift was therefore -0.90 points below the most constrained expectation. That reference had a central simulation envelope of -1.54 to $+0.24$ points. A Monte Carlo uncertainty bracket for its lower edge, -1.559 to -1.523 , contained the observed shift. This comparison does not establish a robust departure from the matched reference. Adverb and modal observations lay inside every reference envelope (Fig. 4; Online Resource 1, Table S14).

Within criterion-negative sources, replication-word incidence increased by 0.360 points in all texts but decreased by 1.170 points in retained texts. The year-and-length-matched expectation was a decrease of 0.520 points. The within-group reversal could therefore arise from the matched composition alone, without establishing residual selection or reversal of the between-group contrast. The decomposition and the retention benchmark answer different accounting questions; neither assigns causal shares.

Frozen scores describe a different conditional outcome

Table 3 reports all three original model-score endpoints under the corrected baseline. Equal-paper means were 0.5004 in criterion-negative sources and 0.5047 in criterion-positive sources. Mean project-aligned changes were positive in both groups, $+0.0085$ and $+0.0054$, respectively. Their small difference therefore does not imply an absence of within-group change. The raw lifetime-score slope contrast was $+0.0002$ per year (-0.0031 to $+0.0036$). These endpoints describe the archived scoring policy, whose semantic limitations remain as defined in Methods.

Table 3 Secondary frozen model-score contrasts, criterion-negative minus criterion-positive. Intervals condition on the archived scoring policy and source reconstruction. Transformed lifetime drift is not raw annual score change.

Endpoint	Negative / positive n	Contrast	95% source interval
Mean score	51 / 45	-0.00421	$[-0.02175, +0.01339]$
Transformed lifetime drift	45 / 40	-0.00052	$[-0.00718, +0.00631]$
Project-aligned change	40 / 32	$+0.00306$	$[-0.02397, +0.02994]$

The 106-source Staunton sensitivity added 22 texts, including 16 Qwen-positive texts. Its score-level contrast was -0.00326 $[-0.02074, +0.01458]$ across 97 eligible sources. This source could not fill three eight-text lifetime bins and had only seven pre-period and nine post-period positive texts. Drift and project-aligned score estimates therefore remained unchanged. It contained no adverb or replication-family hits, and the matched 72-source filter result was unchanged. Online Resource 1 reports broader

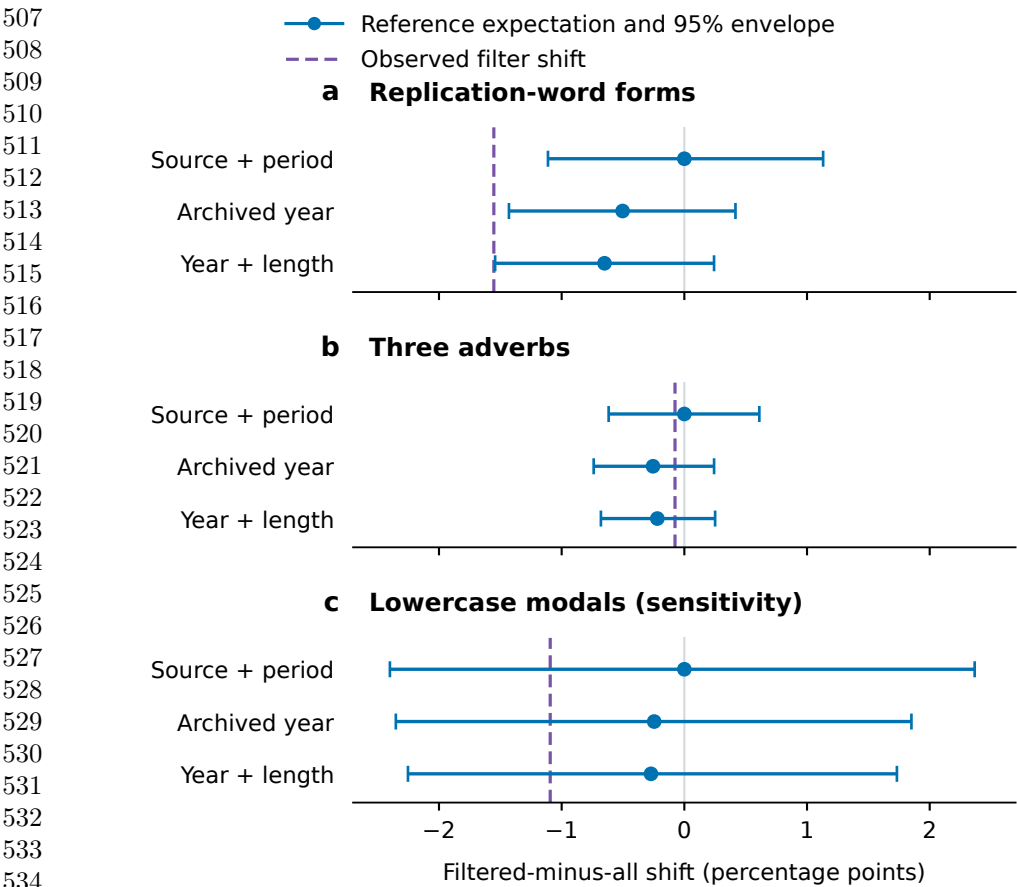


Fig. 4 Observed filter shifts against matched-composition references on the same 72 sources. **a** Replication-word forms. **b** Three adverbs. **c** Lowercase modals. Points are exact marginal expectations; bars are central 95% simulation envelopes from the frozen 19,999 joint draws, and dashed lines are the observed filtered-minus-all shifts. Each scheme preserves 6,650 retained positions among 11,999 texts. Counts are matched within source and period, then more finely by archived year and by year and length. All nine specifications and a common horizontal scale are shown. Envelopes are finite-corpus reference distributions, not source-bootstrap confidence intervals or causal null intervals. In panel a, the observed -1.553 -point shift falls within the Monte Carlo uncertainty bracket for the lower year-and-length envelope edge (-1.559 to -1.523 points); apparent boundary separation is not robust timing, project and source-assignment sensitivities, including the method-sensitive pre-event slope contrast. These analyses establish neither parallel trends nor equivalence between criterion groups.

Discussion

The same source papers yielded different longitudinal descriptions after text selection. Empirical-restatement filtering shifted the replication-word contrast, whereas the adverb and modal families gave weaker evidence. Matching archived year and length reproduced part of the shift and the direction reversal within criterion-negative sources. The observed between-group shift lay close to the most constrained reference boundary, with Monte Carlo uncertainty preventing a sharp separation. An overall retention percentage would not reveal these compositional dependencies.

This result concerns a different stage of measurement from deciding where a citation context begins and ends. Context expansion can recover sentiment missed by sentence-only annotation (Athar and Teufel, 2012), and inconsistent citance boundaries can reduce classification reliability (Taşkın, 2026). Here, text boundaries and filter outputs were held fixed; retaining or excluding those units changed an aggregate comparison over time. The filter could perform its intended empirical-restatement task accurately while still changing the population to which the description applies. Its accuracy, and its coverage of acknowledgment of contrary evidence, remain separate semantic questions.

For comparable longitudinal studies, the practical implication is to report the populations before and after filtering and to repeat key contrasts on the same eligible sources. Composition references can then show whether measured features of selection reproduce the observed pattern. In this case, matching year and length changed the expected replication-word contrast. That finding supports reporting both populations, not treating one reference scheme as a correction that recovers a true causal effect. Literal counts also remain limited: replication has several meanings, modals can be non-epistemic, and three adverbs cover little uncertainty vocabulary.

This sensitivity concerns corpus membership, rather than only the representation of the texts retained. Denny and Spirling (2018) show that preprocessing choices can change results in unsupervised text analysis and propose examining that sensitivity. Our comparison keeps the literal rules fixed and changes the selected population on common source support. Both stages warrant inspection, but matching text composition does not validate the semantics of either the filter or the indicator.

Source-level criteria cannot substitute for claim-level interpretation. Leng (2021) show that different citing communities refer to different findings from a single study. In a distinct setting, Schmidt (2024) find that continued citations to retracted work can serve different informational purposes; a retraction is not the same event as a contradictory replication, but both cases caution against reading citation persistence as uniform endorsement. Citation networks can also propagate distorted accounts (Greenberg, 2009; van der Vet and Nijveen, 2016). Our source-associated criterion therefore does not identify the proposition endorsed by each later text. Independent annotations linking each context to its target claim would be needed to evaluate semantic updating.

The secondary wording score illustrates the same distinction between a reproducible procedure and a validated construct. Its instruction mixes rhetorical commitment with evidential qualification. Agreement with human means improved on reused development items, but those data do not provide independent validation. Successful task-specific evaluations, such as the tweet and news annotation study by Gilardi et al. (2023), show

599 that LLM annotation can be useful; they do not validate a different scientific-wording
600 rubric. Nishikawa and Koshiba (2024) distinguish consistency from predictive accuracy
601 in LLM citation-context classification, while Grimmer and Stewart (2013) emphasize
602 validation for the particular text-analysis task. Our agreement and repeatability checks
603 describe this scoring policy; they cannot establish that its changes measure belief
604 updating or acknowledgment of replication evidence.

605 The study remains an exploratory case analysis. Its focus was chosen post hoc,
606 intervals are pointwise, and two lexical families are sparse. RPP dominates the non-
607 random source sample; omitting it and narrowing the time window yield filter-shift
608 intervals that include zero. Deterministic retrieval and incomplete citing-document
609 and version identifiers leave sampling and dependence uncertainties that exact-text
610 deduplication cannot resolve. Archived years may differ from those of the cited versions.
611 The archive also concerns formally linked citation contexts: Petrovich et al. (2024)
612 show how mention analysis can recover other scholarly connections in historical texts,
613 a population not covered by this retrieval. The external linkage pilot contains no unfa-
614 vorable article-level judgments, so it cannot assess differential coverage of criticism.
615 These constraints limit transfer to other fields and filters even where the numerical
616 analyses can be reproduced.

617 Causal interpretation would require a different design. Egami et al. (2022) identify
618 additional identification and estimation risks when text-based measures are learned
619 and then used for causal inference, and develop a split-sample workflow. Our post hoc
620 reanalysis does not implement that design or isolate exposure to replication evidence.
621 Freezing later analyses and reporting unfavorable specifications make the descriptive
622 sequence inspectable; they do not convert it into preregistered evidence of belief change.

623 The contribution is therefore a measurement account of a particular archive. Holding
624 the source papers fixed makes the effect of text selection visible; retaining excluded
625 texts and matched-composition references helps interpret that difference. Whether later
626 authors recognized contradictory evidence is a further question requiring target-specific
627 semantic evidence.

628

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630

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635 **Competing interests**

636

637 The authors declare no competing interests.

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639 **Author contributions**

640 D.G. and J.L. conceived the study. D.G. implemented retrieval, scoring and analysis.

641 J.L. supervised validation and revision.

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Data and code availability

Online Resource 2 supplies the source-criterion ledger, frozen numerical score inputs, aggregate source-year literal counts, complete sensitivity and decomposition results, protocols, input hashes and executable analysis code. Online Resource 3 supplies the text-free retention-benchmark inputs, joint-pattern counts, full nine-specification results and standalone reproduction code. These supplementary files support arithmetic reproduction without new model calls or human ratings. Earlier project materials are available at <https://github.com/lijiajia3/claim-drift>; the current analyses use the versioned supplementary inputs rather than a changing default branch. Source citation excerpts retain their original publishers' rights. Private annotation-export metadata and participant-level primary-study data are not redistributed. Online Resource 4 supplies the selected article-level labels, text-free gate inputs, complete recovery ledger and descriptive replay for the bounded external-linkage pilot.

Research ethics

The corpus consists of citation contexts from published articles. The manuscript also uses legacy sentence-level ratings, reported under anonymous rater codes. No rater names or contact details are included in the supplementary archive. This study was exempt from ethics review.

Supplementary material

Online Resource 1 provides expanded methods and the complete scientific sensitivity record. Online Resources 2 and 3 provide machine-readable inputs and executable code for the archived-score, literal-indicator, selection-decomposition and retention-benchmark analyses. All three literal families, adverse specifications and historical model diagnostics are retained. Online Resource 4 documents the external provenance and descriptive retention pilot.

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